job_14{1}-Evaluation report

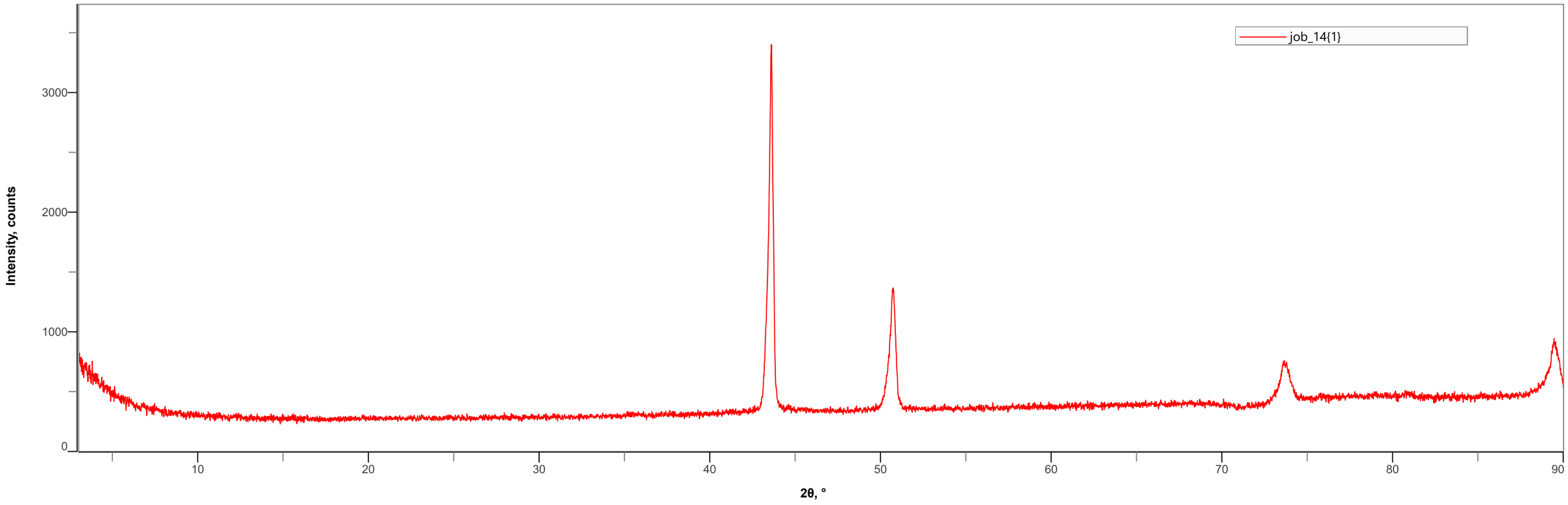
General information

Analysis date	2025-11-27 14:22:39	Measurement start time	2025-11-23 11:54:49
Analyst	Administrator	Operator	Administrator
Sample name	job_14	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\job_14.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	ASC-8	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

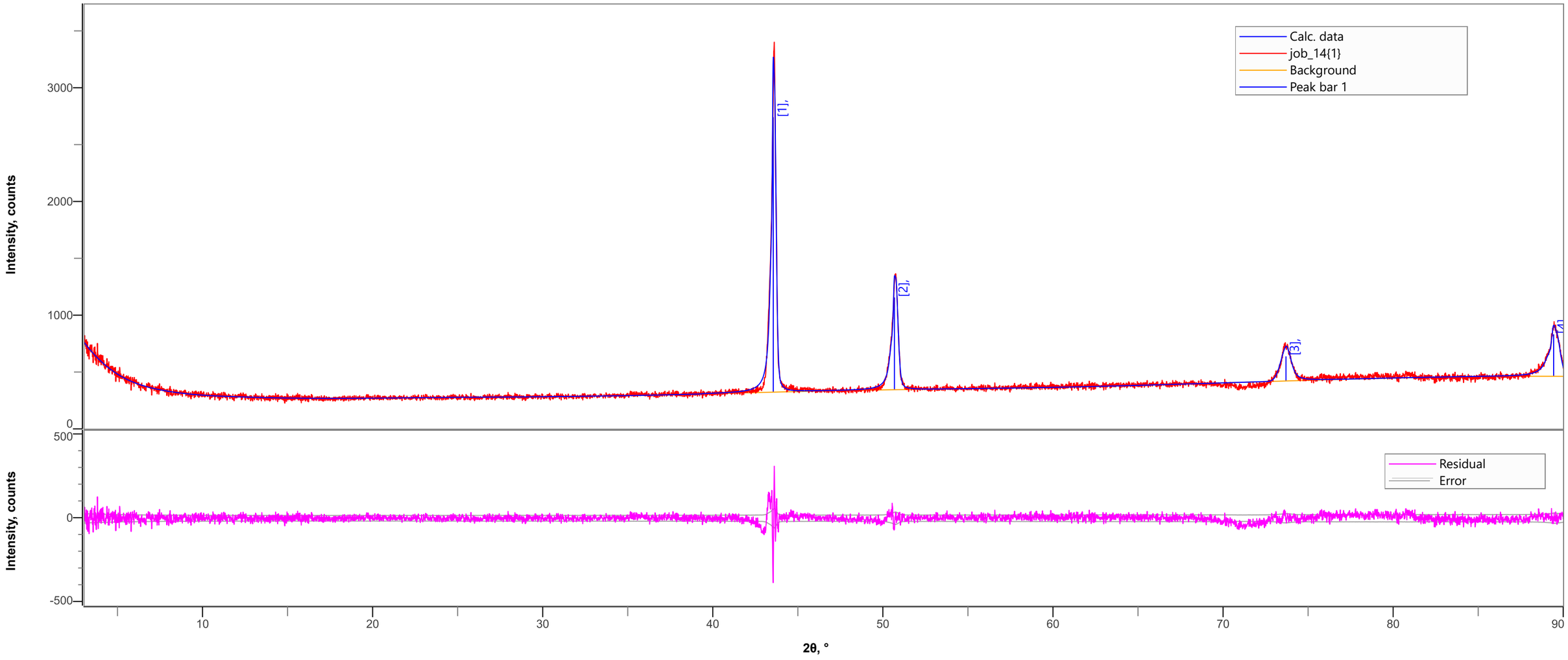
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

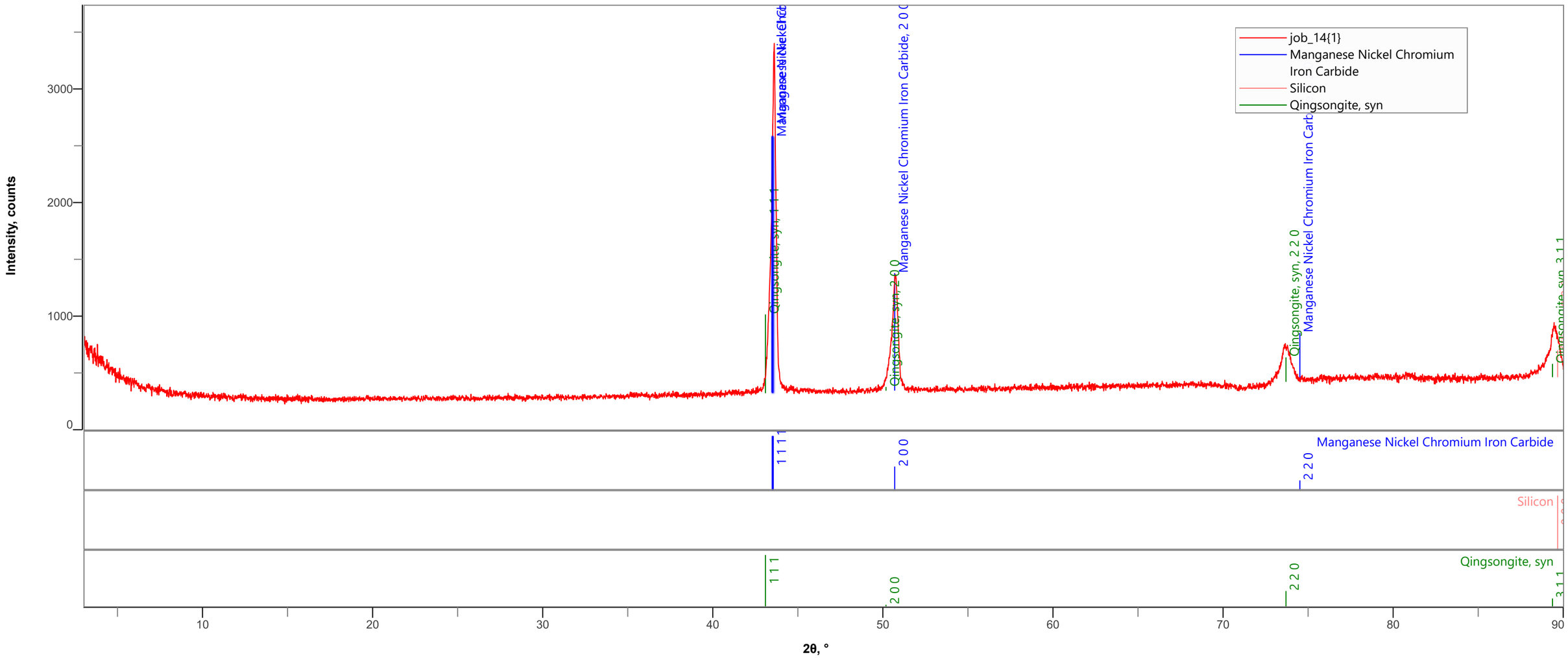
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
Manganese Nickel Chromium Iron Carbide	Cr0.18 Mn0.10 Fe0.56 Ni0.16 C0.0308	0.365	S/M:PDF-5+ 2026	225 : Fm-3m	04-023-5416
Silicon	Si	0.743	S/M:PDF-5+ 2026		00-047-1187
Qingsongite, syn	B N	0.520	S/M:PDF-5+ 2026	216 : F-43m	01-071-6113

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.5300(13)	2.07739(6)	2413(45)	0.199(2)	769(3)	0.319(7)	0.74(2)	1.55(2)	0.382(19)	449(5)	Manganese Nickel Chromium Iron Carbide: 1 1 1	Cr0.18 Mn0.10 Fe0.56 Ni0.16 C0.0308	04-023-5416	100.00	Split Pseudo-Voigt	-	-	-	-	-	
2	50.652(3)	1.80074(11)	810(23)	0.271(5)	352(3)	0.434(16)	0.74(4)	1.55(4)	0.38(4)	339(7)	Manganese Nickel Chromium Iron Carbide: 2 0 0	Cr0.18 Mn0.10 Fe0.56 Ni0.16 C0.0308	04-023-5416	45.73	Split Pseudo-Voigt	-	-	-	-	-	
3	73.66(2)	1.2851(3)	216(8)	0.64(2)	171(4)	0.79(5)	1.6(3)	0.00(11)	0.92(15)	162(5)	Qingsongite, syn: 2 2 0	B N	01-071-6113	22.30	Split Pseudo-Voigt	-	-	-	-	-	
4	89.385(9)	1.09526(8)	368(12)	0.438(14)	259(4)	0.70(3)	0.89(8)	1.55(6)	0.20(12)	266(9)	Silicon,Qingsongite, syn: 3 1 1	Si,B N	00-047-1187,01-071-6113	33.65	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
Manganese Nickel Chromium Iron Carbide	3.60058	3.60058	3.60058	90.000	90.000	90.000
Silicon	0.00000	0.00000	0.00000	90.000	90.000	90.000
Qingsongite, syn	3.63470	3.63470	3.63470	90.000	90.000	90.000

d-I List

Manganese Nickel Chromium Iron Carbide

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.49899	2.07880	1 1 1	100.00
2	50.66535	1.80029	2 0 0	46.09
3	74.47271	1.27300	2 2 0	23.31

Silicon

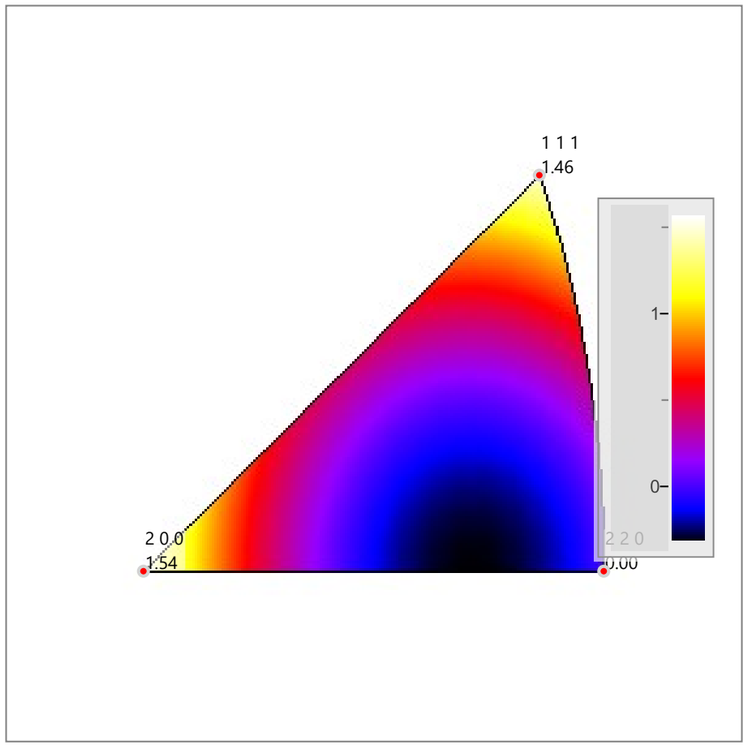
No.	2θ, °	d, Å	h k l	Norm. I.
1	89.61940	1.09300	0 0 0	100.00

Qingsongite, syn

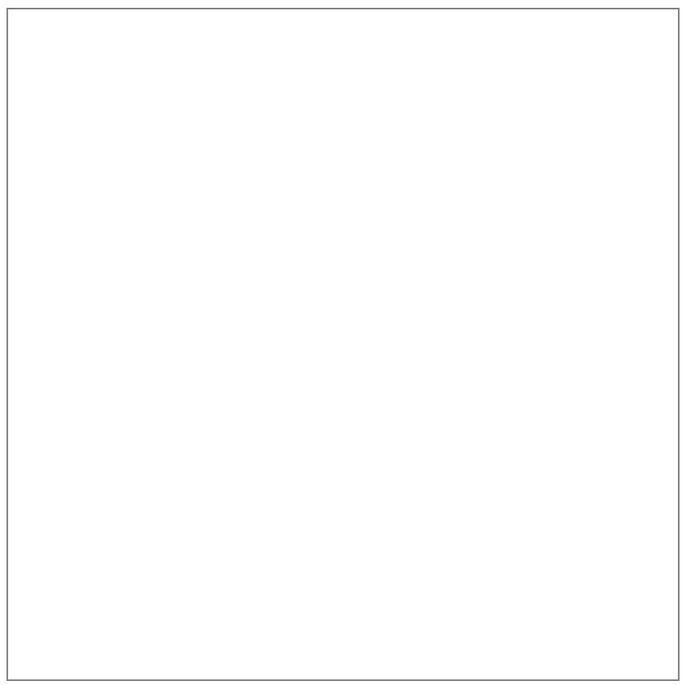
No.	2θ, °	d, Å	h k l	Norm. I.
1	43.07017	2.09849	1 1 1	100.00
2	50.15667	1.81735	2 0 0	4.82
3	73.65736	1.28506	2 2 0	34.17
4	89.31823	1.09590	3 1 1	19.68

Simple Inverse Pole Figure

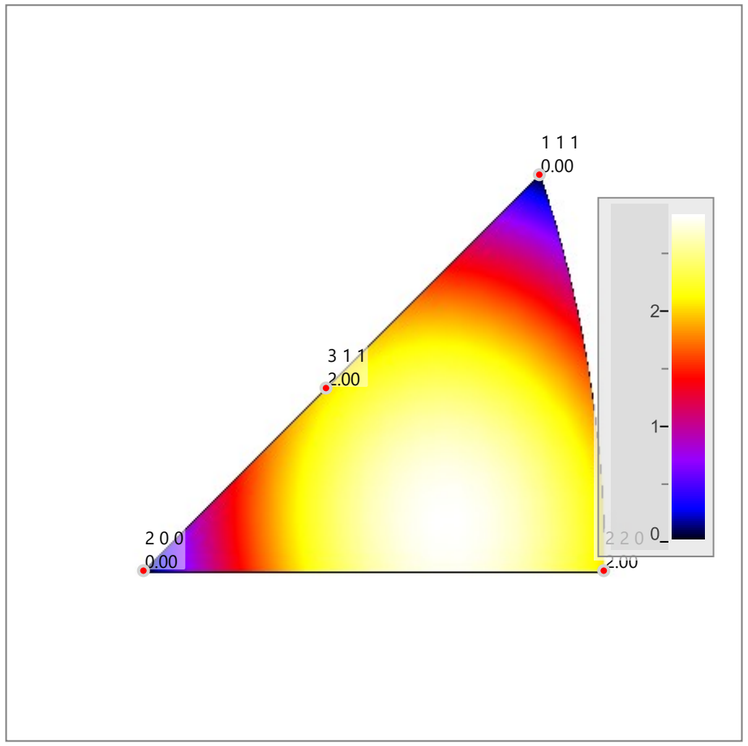
Manganese Nickel Chromium Iron Carbide



Silicon

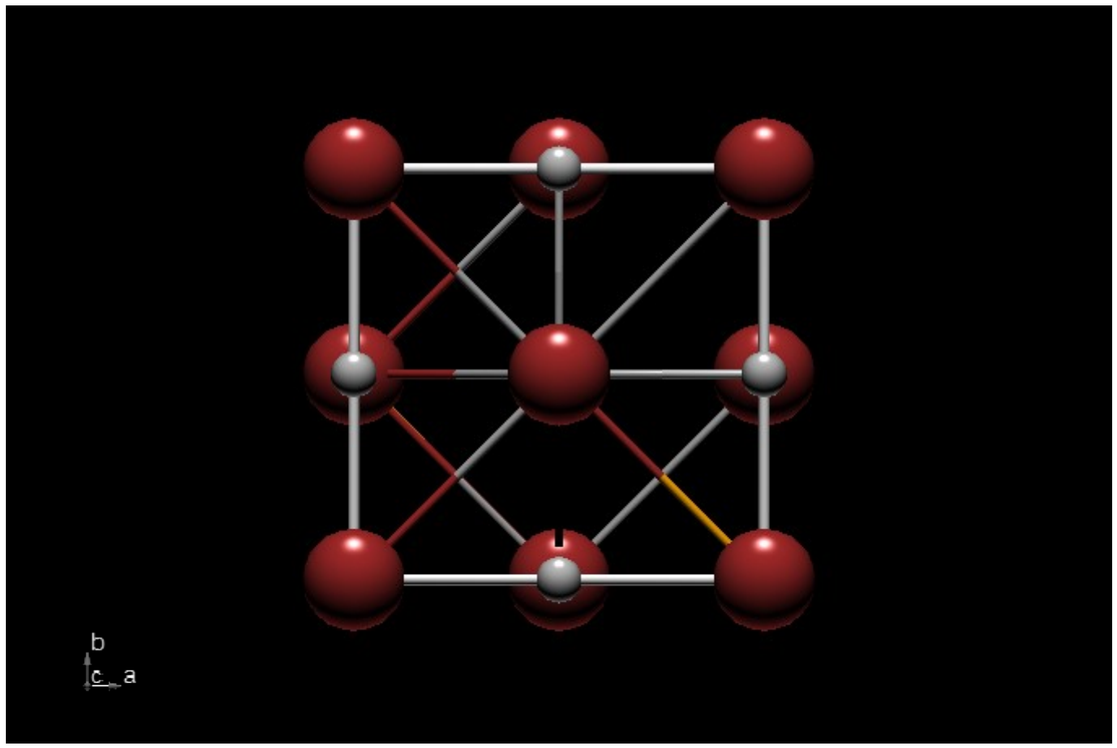


Qingsongite, syn



Crystal Structure

Manganese Nickel Chromium Iron Carbide



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
Fe1(Fe)	0.000000	0.000000	0.000000	0.560	0.500	0	4	0.000	0.000
Cr1(Cr)	0.000000	0.000000	0.000000	0.180	0.500	0	4	0.000	0.000
Ni1(Ni)	0.000000	0.000000	0.000000	0.160	0.500	0	4	0.000	0.000
Mn1(Mn)	0.000000	0.000000	0.000000	0.100	0.500	0	4	0.000	0.000
C1(C)	0.500000	0.500000	0.500000	0.031	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
Ni1-Mn1(1/2+x,y,-1/2+z)	2.54600
Ni1-Mn1(x,1/2+y,1/2+z)	2.54600
Ni1-Mn1(x,-1/2+y,1/2+z)	2.54600
Ni1-Mn1(1/2+x,1/2+y,z)	2.54600
Ni1-Mn1(x,1/2+y,-1/2+z)	2.54600
Ni1-Mn1(-1/2+x,1/2+y,z)	2.54600
Ni1-Mn1(1/2+x,-1/2+y,z)	2.54600
Ni1-Mn1(1/2+x,y,1/2+z)	2.54600

Distance

Bond Def.	Distance, Å
Ni1-Mn1(-1/2+x,y,1/2+z)	2.54600
Ni1-C1(x,-1/2+y,-1/2+z)	1.80029
Ni1-C1(-1+x,-1/2+y,-1/2+z)	1.80029
Ni1-C1(-1/2+x,-1/2+y,z)	1.80029
Ni1-C1(-1/2+x,-1/2+y,-1+z)	1.80029
Ni1-C1(-1/2+x,y,-1/2+z)	1.80029
Ni1-C1(-1/2+x,-1+y,-1/2+z)	1.80029
Fe1-Ni1(1/2+x,y,-1/2+z)	2.54600
Fe1-Ni1(x,1/2+y,-1/2+z)	2.54600
Fe1-Ni1(1/2+x,1/2+y,z)	2.54600
Fe1-Ni1(-1/2+x,1/2+y,z)	2.54600
Fe1-Ni1(1/2+x,-1/2+y,z)	2.54600
Fe1-Ni1(1/2+x,y,1/2+z)	2.54600
Fe1-Ni1(-1/2+x,y,1/2+z)	2.54600
Fe1-Ni1(x,1/2+y,1/2+z)	2.54600
Fe1-Ni1(x,-1/2+y,1/2+z)	2.54600
Fe1-Fe1(1/2+x,-1/2+y,z)	2.54600
Fe1-Fe1(1/2+x,y,1/2+z)	2.54600
Fe1-Fe1(1/2+x,y,-1/2+z)	2.54600
Fe1-Fe1(x,1/2+y,1/2+z)	2.54600
Fe1-Fe1(-1/2+x,y,1/2+z)	2.54600
Fe1-Fe1(x,-1/2+y,1/2+z)	2.54600
Fe1-Fe1(x,1/2+y,-1/2+z)	2.54600
Fe1-Fe1(1/2+x,1/2+y,z)	2.54600
Fe1-Fe1(-1/2+x,1/2+y,z)	2.54600
Fe1-Mn1(1/2+x,y,1/2+z)	2.54600
Fe1-Mn1(-1/2+x,y,1/2+z)	2.54600
Fe1-Mn1(1/2+x,y,-1/2+z)	2.54600
Fe1-Mn1(x,1/2+y,1/2+z)	2.54600
Fe1-Mn1(x,-1/2+y,1/2+z)	2.54600
Fe1-Mn1(x,1/2+y,-1/2+z)	2.54600
Fe1-Mn1(1/2+x,1/2+y,z)	2.54600
Fe1-Mn1(-1/2+x,1/2+y,z)	2.54600
Fe1-Mn1(1/2+x,-1/2+y,z)	2.54600
Fe1-Cr1(-1/2+x,1/2+y,z)	2.54600
Fe1-Cr1(1/2+x,y,-1/2+z)	2.54600
Fe1-Cr1(-1/2+x,y,1/2+z)	2.54600
Fe1-Cr1(1/2+x,y,1/2+z)	2.54600
Fe1-Cr1(1/2+x,-1/2+y,z)	2.54600
Fe1-Cr1(1/2+x,1/2+y,z)	2.54600
Fe1-Cr1(x,1/2+y,-1/2+z)	2.54600

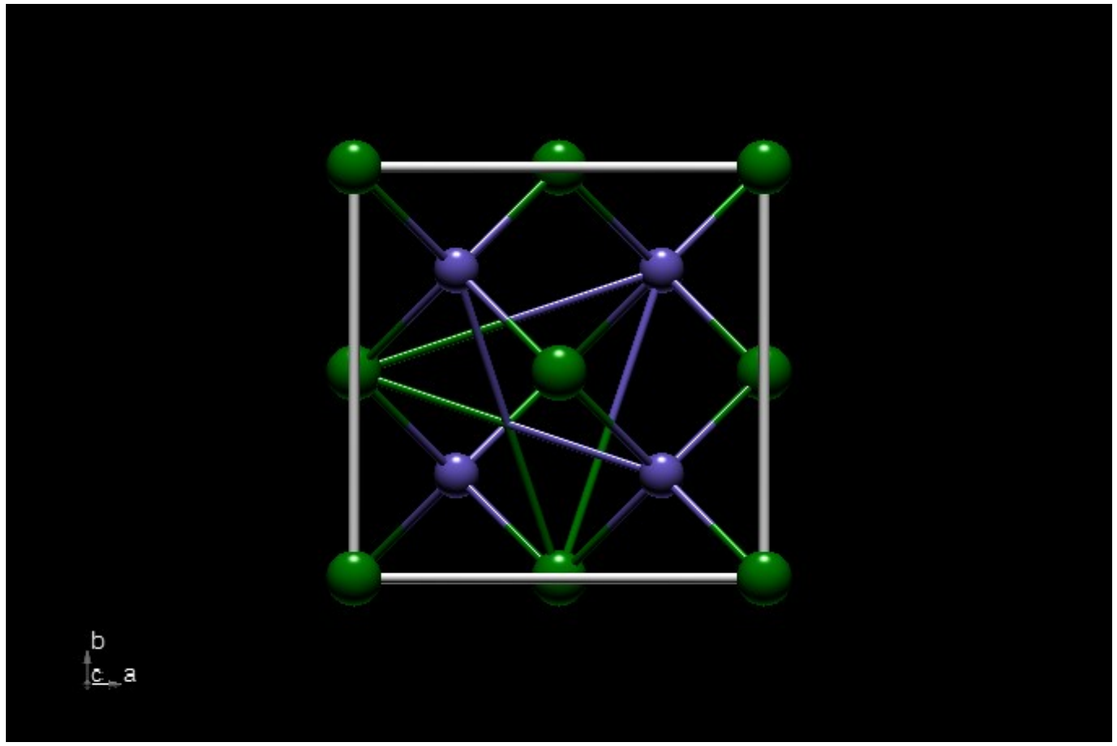
Distance

Bond Def.	Distance, Å
Fe1-Cr1(x,-1/2+y,1/2+z)	2.54600
Fe1-Cr1(x,1/2+y,1/2+z)	2.54600
Fe1-C1(x,-1/2+y,-1/2+z)	1.80029
Fe1-C1(-1+x,-1/2+y,-1/2+z)	1.80029
Fe1-C1(-1/2+x,-1/2+y,z)	1.80029
Fe1-C1(-1/2+x,-1/2+y,-1+z)	1.80029
Fe1-C1(-1/2+x,y,-1/2+z)	1.80029
Fe1-C1(-1/2+x,-1+y,-1/2+z)	1.80029
Mn1-Mn1(x,-1/2+y,1/2+z)	2.54600
Mn1-Mn1(x,1/2+y,1/2+z)	2.54600
Mn1-Mn1(x,1/2+y,-1/2+z)	2.54600
Mn1-Mn1(1/2+x,1/2+y,z)	2.54600
Mn1-Mn1(-1/2+x,1/2+y,z)	2.54600
Mn1-Mn1(1/2+x,-1/2+y,z)	2.54600
Mn1-Mn1(1/2+x,y,1/2+z)	2.54600
Mn1-Mn1(-1/2+x,y,1/2+z)	2.54600
Mn1-Mn1(1/2+x,y,-1/2+z)	2.54600
Mn1-C1(-1/2+x,-1+y,-1/2+z)	1.80029
Mn1-C1(-1/2+x,-1/2+y,z)	1.80029
Mn1-C1(-1+x,-1/2+y,-1/2+z)	1.80029
Mn1-C1(x,-1/2+y,-1/2+z)	1.80029
Mn1-C1(-1/2+x,y,-1/2+z)	1.80029
Mn1-C1(-1/2+x,-1/2+y,-1+z)	1.80029
Cr1-Ni1(1/2+x,y,1/2+z)	2.54600
Cr1-Ni1(-1/2+x,y,1/2+z)	2.54600
Cr1-Ni1(x,-1/2+y,1/2+z)	2.54600
Cr1-Ni1(1/2+x,y,-1/2+z)	2.54600
Cr1-Ni1(x,1/2+y,-1/2+z)	2.54600
Cr1-Ni1(1/2+x,1/2+y,z)	2.54600
Cr1-Ni1(-1/2+x,1/2+y,z)	2.54600
Cr1-Ni1(1/2+x,-1/2+y,z)	2.54600
Cr1-Ni1(x,1/2+y,1/2+z)	2.54600
Cr1-Mn1(-1/2+x,y,1/2+z)	2.54600
Cr1-Mn1(1/2+x,y,-1/2+z)	2.54600
Cr1-Mn1(x,1/2+y,1/2+z)	2.54600
Cr1-Mn1(x,-1/2+y,1/2+z)	2.54600
Cr1-Mn1(x,1/2+y,-1/2+z)	2.54600
Cr1-Mn1(1/2+x,1/2+y,z)	2.54600
Cr1-Mn1(-1/2+x,1/2+y,z)	2.54600
Cr1-Mn1(1/2+x,-1/2+y,z)	2.54600
Cr1-Mn1(1/2+x,y,1/2+z)	2.54600

Distance

Bond Def.	Distance, Å
Cr1-Cr1(1/2+x,y,-1/2+z)	2.54600
Cr1-Cr1(-1/2+x,y,1/2+z)	2.54600
Cr1-Cr1(1/2+x,y,1/2+z)	2.54600
Cr1-Cr1(1/2+x,-1/2+y,z)	2.54600
Cr1-Cr1(x,1/2+y,1/2+z)	2.54600
Cr1-Cr1(x,-1/2+y,1/2+z)	2.54600
Cr1-Cr1(x,1/2+y,-1/2+z)	2.54600
Cr1-Cr1(1/2+x,1/2+y,z)	2.54600
Cr1-Cr1(-1/2+x,1/2+y,z)	2.54600
Cr1-C1(x,-1/2+y,-1/2+z)	1.80029
Cr1-C1(-1+x,-1/2+y,-1/2+z)	1.80029
Cr1-C1(-1/2+x,-1/2+y,z)	1.80029
Cr1-C1(-1/2+x,-1/2+y,-1+z)	1.80029
Cr1-C1(-1/2+x,y,-1/2+z)	1.80029
Cr1-C1(-1/2+x,-1+y,-1/2+z)	1.80029

Qingsongite, syn



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
B1(B)	0.000000	0.000000	0.000000	1.000	0.500	0	4	0.000	0.000
N1(N)	0.250000	0.250000	0.250000	1.000	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
B1-N1	1.57387
B1-N1(x,-y,-z)	1.57387
B1-N1(-x,y,-z)	1.57387
B1-N1(-x,-y,z)	1.57387